

Field Oriented Control (FOC) for a 3-Phase Induction Motor

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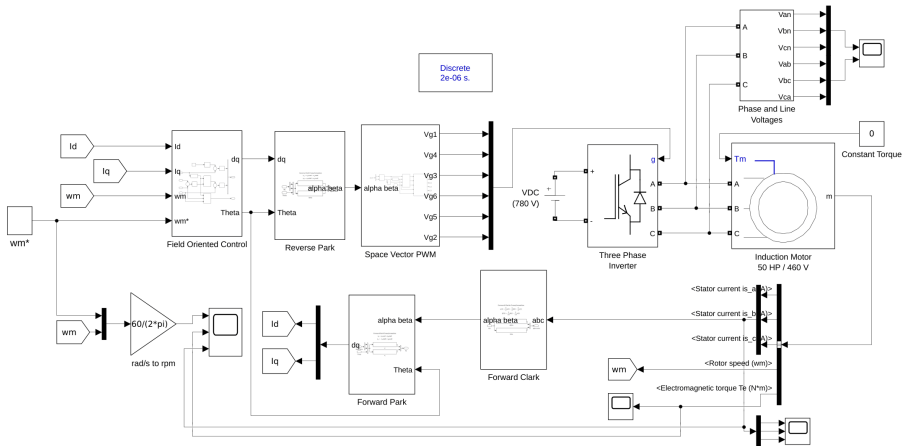
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The Evolution of AC Motor Drives

- **The Legacy of Scalar Control:** Traditional Constant Volts-per-Hertz (V/f) control governs only magnitude and frequency. It ignores spatial magnetic orientation, leading to sluggish transient responses and coupled torque/flux dynamics.
- **The FOC Paradigm Shift:** FOC (Vector Control) emulates the highly desirable, decoupled control dynamics of a separately excited DC motor within an AC machine.
- **Project Objective:** To systematically deconstruct and simulate an Indirect Field Oriented Control (IFOC) architecture for a 50 HP Squirrel-Cage Induction Motor using MATLAB/Simulink.

System Architecture Overview

Top-Level IFOC Block Diagram



- **Outer Loop:** Speed regulation PI controller generating the torque command (T_e^*).
- **Inner Loop:** High-bandwidth PI current regulators (i_d, i_q).

The Induction Motor Plant Parameters

Equivalent Circuit Specifications

- **Motor Ratings:** 50 HP (37.3 kW), 460 V, 50 Hz, 4-Pole (1500 RPM).
- **Electrical Parameters:**
 - Stator Res. (R_s): 0.087 Ω
 - Rotor Res. (R_r): 0.228 Ω
 - Mag. Ind. (L_m): 34.7 mH

Derived Self-Inductances:

$$L_s = L_{ls} + L_m = 34.8 \text{ mH}$$

$$L_r = L_{lr} + L_m = 35.5 \text{ mH}$$

Rotor Time Constant:

$$\tau_r = \frac{L_r}{R_r} = 0.1557 \text{ s}$$

Understanding the Mechanical Transfer Function

- **Mechanical Equation of Motion:**

$$T_e - T_L = J \frac{d\omega_m}{dt} + F\omega_m$$

- **Mechanical Parameters:** Inertia (J) = $0.662 \text{ kg} \cdot \text{m}^2$, Friction (F) = 0.1 Nms/rad .
- **Laplace Domain Transfer Function:**

$$\frac{\omega_m(s)}{T_e(s)} = \frac{1}{0.662s + 0.1}$$

- **Significance:** The high inertia acts as a natural low-pass filter, naturally attenuating high-frequency torque ripples generated by the SVPWM inverter.

Coordinate Transformations - The Core Concept

Resolving the AC Control Dilemma

- **The Problem:** Standard PI controllers suffer from finite bandwidth limits and cannot track time-varying sinusoidal AC signals without steady-state amplitude and phase errors.
- **The Solution:** Mathematical spatial projections.
- By transforming variables into a synchronously rotating reference frame ($d - q$), FOC mathematically forces these alternating signals to appear as trackable, flat DC quantities.
- This allows our standard PI controllers to achieve zero steady-state error.

Forward Clarke Transform

Stationary $3\phi \rightarrow 2\phi$ Projection

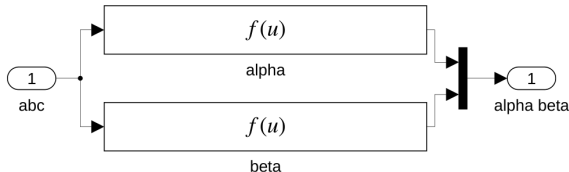
- Simplifies the 3-phase system by projecting currents (i_a, i_b, i_c) onto a 2D stationary orthogonal frame ($\alpha - \beta$).
- Uses the amplitude-invariant matrix to preserve peak current amplitudes.

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$

Forward Clarke Transformation

$$\alpha(t) = \frac{2}{3}a(t) - \frac{1}{3}b(t) - \frac{1}{3}c(t)$$

$$\beta(t) = \frac{1}{\sqrt{3}}b(t) - \frac{1}{\sqrt{3}}c(t)$$

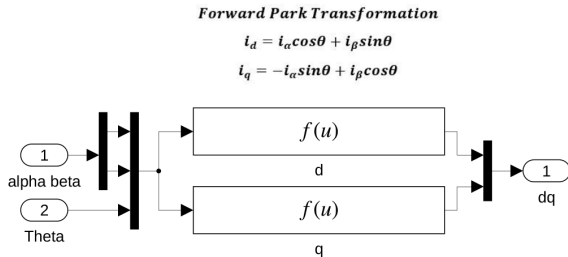


Forward Park Transform

Stationary $\alpha - \beta \rightarrow$ Rotating $d - q$ Projection

- Projects the stationary $\alpha - \beta$ frame onto a reference frame spinning at the electrical angular velocity of the rotor magnetic flux (θ_e).
- Physically locks the observer's perspective to the spinning magnetic field, fully demodulating AC currents into DC.

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos \theta_e & \sin \theta_e \\ -\sin \theta_e & \cos \theta_e \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}$$

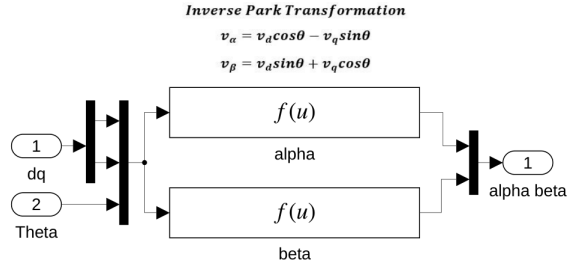


Reverse Park Transform

Coordinate De-rotation for Actuation

- PI current regulators process DC error signals and output decoupling voltage commands (v_d^* and v_q^*).
- Because the physical inverter operates in the stationary frame, these vectors must be de-rotated.

$$\begin{bmatrix} v_\alpha^* \\ v_\beta^* \end{bmatrix} = \begin{bmatrix} \cos \theta_e & -\sin \theta_e \\ \sin \theta_e & \cos \theta_e \end{bmatrix} \begin{bmatrix} v_d^* \\ v_q^* \end{bmatrix}$$



Flux Control (Direct-Axis Decoupling)

Independent Magnetization Control

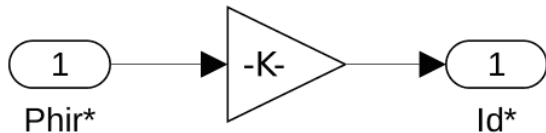
- The d-axis current (i_{ds}) strictly governs magnetization, analogous to field excitation in a DC motor.
- Under steady-state conditions ($\frac{d\psi_r}{dt} = 0$), the relationship simplifies to:

$$i_{ds}^* = \frac{\psi_r^*}{L_m}$$

- For a nominal flux of 0.96 Wb and $L_m = 0.0347$ H, the reference current is firmly regulated to 27.67 **A**.

Reference I_d Calculation

$$I_d = \frac{\varphi_r}{L_m}$$



Torque Control (Quadrature-Axis Decoupling)

Independent Active Force Control

- The q-axis current (i_{qs}) is maintained precisely at a 90° angle to the flux axis ($\psi_{qr} = 0$).
- Decoupled Torque Equation:**

$$T_e = \frac{3}{2} \left(\frac{P}{2} \right) \frac{L_m}{L_r} \psi_r i_{qs}$$

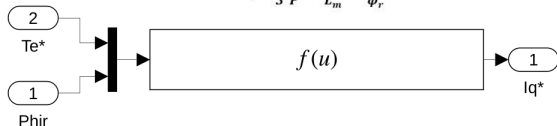
- Isolating i_{qs}^* yields a direct torque constant:

$$T_e = 2.8152 \cdot i_{qs} \text{ N} \cdot \text{m}$$

Reference I_q Calculation

$$T_e = \frac{3}{2} \frac{P}{2} \frac{L_m}{L_{lr} + L_m} I_q \varphi_r$$

$$I_q = \frac{2}{3} \frac{2}{P} \frac{L_{lr} + L_m}{L_m} \frac{T_e}{\varphi_r}$$



Preventing Division-by-Zero Singularities

- A critical engineering consideration occurs during motor startup ($t = 0$) when the machine is completely unmagnetized ($\psi_r \approx 0$).
- Native execution of the i_{qs}^* equation would cause a division-by-zero singularity, crashing the DSP or driving PI regulators into windup.
- **Solution:** The Simulink algorithm introduces a numerical perturbation ($\epsilon = 10^{-3}$) to the flux denominator:

$$i_{qs}^* \propto \frac{T_e^*}{\psi_r + 10^{-3}}$$

- Keeps the starting current extremely high, but safely bounded.

Slip and Flux Angle (θ_e) Synthesis

The Core of Indirect FOC (IFOC)

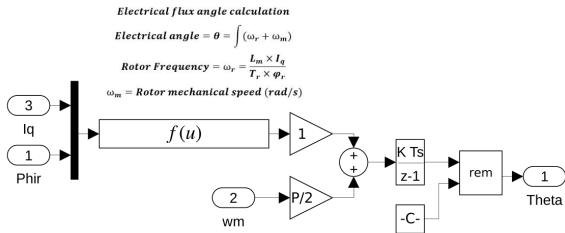
- IFOC mathematically synthesizes the flux angle using a feedforward slip calculation, avoiding fragile air-gap sensors.

- Required Slip Frequency:**

$$\omega_{sl} = \frac{L_m}{\tau_r} \cdot \frac{i_{qs}^*}{\psi_r^*}$$

- Synchronous Speed:**

$$\omega_e = \omega_r + \omega_{sl}$$



Discrete Integration for Angle (θ_e)

Digital Implementation Constraints

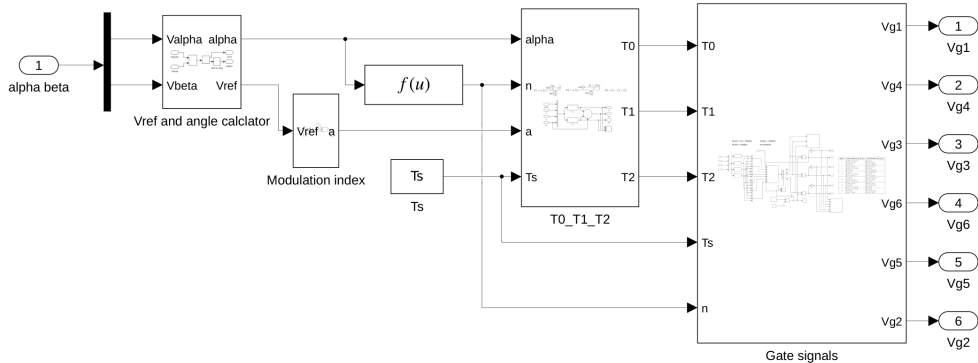
- The instantaneous rotor flux angle (θ_e) must be calculated by integrating the synchronous speed continuously.
- Executed at the model's fundamental $2\mu\text{s}$ discrete sample time using Forward Euler:

$$\theta_e[k] = (\theta_e[k-1] + \omega_e[k] \cdot T_s) \pmod{2\pi}$$

- The modulo 2π operator wraps the angle within $[0, 2\pi]$, preventing numerical overflow in DSP registers during continuous operation.

Space Vector PWM (SVPWM)

Connecting Mathematics to Power Electronics



- Translates continuous v_{α}^* and v_{β}^* commands into discrete transistor switching signals for the 780 V DC-link inverter.
- **Why SVPWM?** Injects zero-sequence voltages, enhancing DC Bus Utilization by $\approx 15.47\%$ and drastically minimizing Total Harmonic Distortion (THD).

SVPWM Algorithm - Reference Synthesis

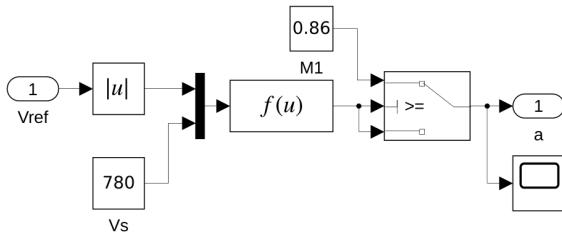
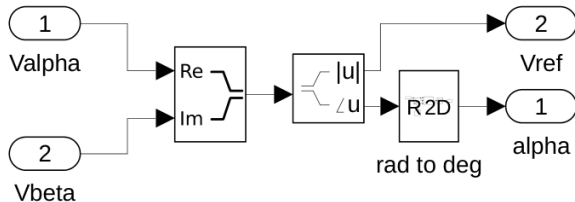
Calculating Magnitude and Modulation Index

- Converts stationary commands into a rotating polar vector:

$$|\vec{V}_{ref}| = \sqrt{(v_{\alpha}^*)^2 + (v_{\beta}^*)^2}$$

$$\theta_{ref} = \arctan\left(\frac{v_{\beta}^*}{v_{\alpha}^*}\right)$$

- Modulation Index (a):** Bounded by a saturation switch (≤ 0.866) to prevent deep overmodulation and distortion.



SVPWM Algorithm - Dwell Times

Applying the Volt-Second Balance Principle

- Synthesizes the target vector by time-averaging active vectors (T_1 , T_2) and the zero vector (T_0).

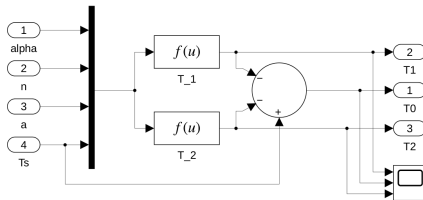
- Equations:**

$$T_1 = a \cdot T_s \frac{\sin\left(\frac{n\pi}{3} - \alpha\right)}{\sin\left(\frac{\pi}{3}\right)}$$

$$T_2 = a \cdot T_s \frac{\sin\left(\alpha - \frac{(n-1)\pi}{3}\right)}{\sin\left(\frac{\pi}{3}\right)}$$

$$T_0 = T_s - T_1 - T_2$$

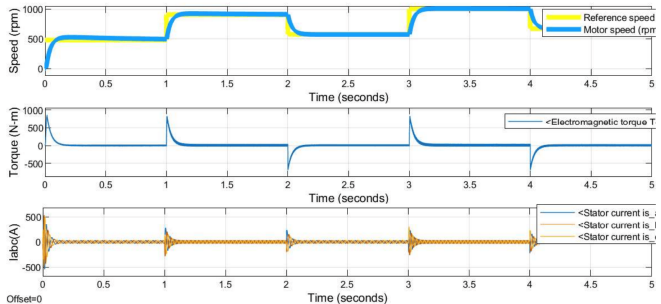
$$T_1 = a \cdot T_s \frac{\sin\left(\frac{n\pi}{3} - \alpha\right)}{\sin\left(\frac{\pi}{3}\right)} \quad T_2 = a \cdot T_s \frac{\sin\left(\alpha - \frac{(n-1)\pi}{3}\right)}{\sin\left(\frac{\pi}{3}\right)} \quad T_0 = T_s - T_1 - T_2$$



Simulation Results: Transient Tracking

Validating Dynamic Response

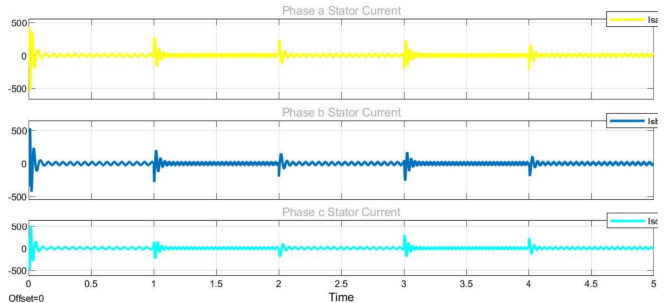
- Motor tracks aggressive staircase speed references flawlessly.
- Decoupling verified: Torque (T_e) acts directly and instantaneously to speed errors.
- Peaks near $\approx 830 \text{ N} \cdot \text{m}$ during acceleration; drops to $\approx -680 \text{ N} \cdot \text{m}$ for regenerative braking.



Simulation Results: Current Regulation

Validating the Inner Control Loops

- Steady-state phase currents drop to a tiny magnetizing envelope.
- During speed transients, current envelope flares symmetrically to peak amplitudes near 500 A.
- Accurate translation of PI torque commands without unstable overcurrent limits.



Conclusion & Future Scope

Conclusion:

- Validated a functionally complete "digital twin" of an industrial IFOC drive.
- Demonstrated how non-linear, multi-variable AC machines can be transformed into decoupled, linear systems yielding DC-like performance.

Future Scope:

- **Sensorless Control:** Implement Model Reference Adaptive Systems (MRAS) to eliminate the physical speed encoder.
- **Online Parameter Identification:** Dynamically estimate rotor resistance (R_r) to prevent vector detuning caused by thermal heating.
- **Field Weakening:** Actively attenuate i_{ds} to push the machine into the constant-power region above base speed.